

• V1.0.0 — STANDARD LIBRARY ONLY

Principal Strategist

A 90-day hard-mode FinTech Arbitrage Specialist simulator — *zero external dependencies, maximum rigor.*

Python 3.8+

No pip install

CMA / SOCPA / IFRS

Double-Entry Ledger

AST Code Audit

Live Web Dashboard

See Screenshots

Explore Features

● ● ● NovaCap Principal Strategist – Arbitrage Academy

PRINCIPAL STRATEGIST

NovaCap Financial Technologies Ltd. – 90-Day Arbitrage Protocol

WELCOME, TRAINEE.

You are enrolled in the NovaCap 90-Day FinTech Arbitrage Protocol.

Standards enforced: CMA (Certified Management Accountant)

SOCPA (Saudi CPA)

IFRS (International Financial Reporting)

OBSERVATION: This is not a course. This is a simulation of

OBSERVATION: the most demanding trading desk in the region.

DIRECTIVE: Type "start" to begin Day 1.

Type "help" for available commands.

strategist> start

[✓] Starting mock exchange server...

[✓] Exchange operational on :8080

[✓] Day 1 workspace ready

Scroll to explore



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[Features](#)[Screenshots](#)[Architecture](#)[Pricing](#)[Get Started](#)[Launch Dashboard →](#)**3,961**

LINES OF PYTHON

11INTERCONNECTED
MODULES**90**

TRAINING DAYS

0

PIP DEPENDENCIES

★ CORE MODULES

Everything in *one file*.

A single Python script (3,961 lines) containing every module — no imports, no pip install, no setup. Written for Python 3.8+ on any OS.

**Mock Exchange Server**

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SQLite Ledger Engine

Double-entry bookkeeping with 9 seeded accounts (cash, crypto, revenue, fees). Tracks every trade, verifies debits = credits, and calculates P&L per symbol — all in SQLite with zero setup.

[Double-Entry](#)[SQLite](#)[P&L](#)

AST Static Code Auditor

Parses your solution with Python's ast module. Detects: missing try-except on network calls, $O(n^2)$ nested loops, eval/exec, division by zero, missing type annotations, infinite while loops, and global state pollution.

[AST](#)[Static Analysis](#)[Security](#)

Evaluation Engine

EOD pipeline: reads user code, runs AST audit, executes hidden test suite (subprocess), verifies SQLite ledger integrity, calculates score (0–100). Critical violations = automatic 0. Must score 100 to advance.

[Subprocess](#)[Scoring](#)[Validation](#)

Web UI Dashboard

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Launch Dashboard →

Threaded HTTP server on :8081 serving a full dark-themed dashboard. Live ticker feed, code editor with syntax highlighting, save/run, evaluation results history, ledger explorer, AST audit viewer — all with auto-polling.

HTML/CSS/JSREST APIDark Theme



Workspace Generator

Generates per-day workspace folders with DAILY_BRIEF.md, boilerplate .py files, and hidden test suites. Days 1–3 have hand-crafted curriculums; days 4–90 use procedural generation.

File GenerationCurriculum

SCREENSHOTS

See it in action.

Live captures from the running application — mock exchange data, Web UI panels, CLI output, and ledger views.

CURRENT DAY

1 / 90

PHASE

Data Ingestion

DAYS PASSED

0

EXCHANGE

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● ONLINE

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Launch Dashboard →

BTC/USD\$67,306.92ETH/USD\$3,126.42BTC/USDT\$68,188.97EUR/USD\$1.09

Start DayEODRunAdvance

Web UI Dashboard

Live market feed, stat cards (current day/phase/score), action buttons, and auto-polling ticker every 3 seconds. Dark theme inspired by trading terminals.

```
{
  "symbol": "BTC/USD",
  "price": 67306.92,
  "change_24h_pct": -4.07,
  "volume_24h": 5739.1,
  "timestamp": "2026-06-18T22:53:26.463304"
}
```

GET /v1/ticker?symbol=BTC/USD | GET /v1/orderbook?symbol=ETH/USDT | POST /v1/execute | GET /v1/health

Mock Exchange REST API

Threaded HTTP server on :8080 with order books, tickers, trade execution, and health checks. Every endpoint returns live JSON with realistic market data.

▲ BIDS

3,142.8940.45

3,136.6033.69

3,130.3222.72

3,124.0336.99

3,117.7513.08

▼ ASKS

3,151.1640.45

3,157.4633.69

3,163.7622.72

3,170.0636.99

3,176.3713.08

Spread: 8.27 — Mid: 3,146.82

ETH/USDT Order Book — 5 Levels

Live order book from the mock exchange showing bids (buyers) and asks (sellers) with price levels and quantities. Spread and mid-price calculated automatically.

CODE	NAME	TYPE	BALANCE
1000	Cash - USD	asset	100,000.00
1100	BTC Wallet	asset	0.50
1200	ETH Wallet	asset	0.00
1300	USDT Wallet	asset	50,000.00
4000	Revenue - Trading	revenue	1,245.30
5000	Fees Paid	expense	33.72

✓ Double-entry balanced — Trades logged: 1

SQLite Ledger — Double-Entry Bookkeeping

9 seeded accounts across assets, revenue, and expense types. Every trade creates balanced debit/credit entries. P&L calculated per symbol with SOCPA-compliant audit trails.

```
1  # AST Auditor - detected issues
• CRITICAL L5  Missing try-except on urllib.request.urlopen()
• WARNING L11  O(n²) nested loop: for i in range(len(prices))
• INFO L4      Missing type annotation for parameter 'url'
• INFO L8      Missing return type annotation
• CRITICAL L3  Use of eval() detected - security risk
--- Complexity Score: 14 - 4 critical, 1 warning, 2 info
```

AST Static Code Audit

Real-time code analysis using Python's ast module. 9 check categories: network exception handling, nested loops, eval/exec, type annotations, division by zero, infinite loops, and global state.

```
1  # Day 1: Data Ingestion & SMA Calculation
2  import json
```

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Launch Dashboard →

```
3 import sqlite3
4
5 from typing import List, Optional, Tuple
6
7 def fetch_orderbook(url: str = EXCHANGE_URL) -> Optional[dict]:
8     """Fetch order book from mock exchange."""
9     # TODO: Implement with proper error handling
10    pass
11
12 def calculate_sma(prices: List[float], window: int) -> List[Optional[float]]:
13     """Calculate Simple Moving Average."""
14     # TODO: Implement
15    pass
```

Code Editor — Day 1 Boilerplate

Built-in code editor with save/run. Day 1 task: fetch BTC/USD order book from mock exchange, calculate SMA(5) and SMA(20), log to SQLite. All boilerplate includes type annotations and error handling stubs.

⚙ ARCHITECTURE

Single-file, multi-threaded.

All modules live in one 3,961-line Python file. Threaded HTTP servers run the exchange and web UI in the background while the CLI processes commands.



CLI

cmd.Cmd interface with 10 commands



Web UI

Threaded HTTP on :8081



Mock Exchange

Threaded HTTP on :8080



AST Auditor



Ledger Engine



Evaluator

Static code analysis via ast

Double-entry SQLite bookkeeping

Subprocess tests + scoring

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Workspace Gen

Per-day file generation



State Manager

JSON persistence & day tracking

DATA FLOW

CLI / Web UI → Workspace Gen → User writes code → EOD: AST Audit → Test Suite →

Ledger Verify → Score

⚡ TECH STACK

Standard library only.

No package manager. No requirements.txt. No node_modules. Built with what ships with Python.



http.server / socketserver

Threaded HTTP servers for exchange & web UI



ast

file:///C:/Downloads/ME/career-roadmap-app/Job/promo.html

8/12

Abstract Syntax Tree parsing for code audits

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sqlite3

Embedded database for double-entry ledger



threading

Background daemon threads for both servers



urllib / json

HTTP client for exchange API calls + JSON parsing



subprocess

Isolated test execution in child processes

★ LICENSING

One file. Four tiers.

Choose the license that fits your use case. All tiers include the same software, but the differences are in what you're allowed to do with it.

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- ✓ Dedicated engineering SLA
- ✓ Custom module development available

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What's the refund policy?



Can I buy now and upgrade later?



What payment methods do you accept?



✓ **30-Day Money-Back Guarantee.** Download the script, run it, and if it doesn't do what we say, get a full refund. No questions asked.

See it in action before you buy.

The Principal Strategist is live on localhost:8081 — start Day 1, write your solution, and face the EOD evaluation. No license key required to try.

▶ Launch Dashboard

View Pricing ↓

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